

Sports-Medical Supervision of Competition Climbers and Climbing Competitions

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19.1 Introduction

In the advent of the inclusion of climbing into the Olympic programs of Tokyo and Paris, the sports-medical supervision of the climbing team has become more. We have already been conducting these exams for the German Alpine club for over 20 years, having performed Olympic-standard examinations long before climbing was on the Olympics list. In other countries, these examinations are newer and need to be organized by the national federations and Olympic centers. As we also take care of regional and state teams, a step concept was developed, in which we start with physical examinations, history, finger ultrasound, rest ECG, and a laboratory test with the lower teams and increase the extent and number of examinations up to the Olympic levels. Adolescent climbers need to be evaluated especially carefully because of their high risk for epiphyseal growth plate fractures [1]. Thus, long-term damage can be avoided and, in an injury, early intervention is possible. The general stages of children's development and maturation are monitored, and early signs of malnutrition must be detected. In the following sections, we discuss the most important components of the team examinations and competition care as well as the most important pathologies.

19.2 Climbing Team Examinations

The yearly team examination is an important tool for prevention and early detection of orthopedic and/or medical conditions. Reversible conditions, such as restricted range of motion in the fingers or growth plate fractures, can be detected early and counteracted. Additionally, the general stages of development of the children and adolescents must be considered. Only then is it possible to detect possible malnutrition, which is a frequent problem in young ambitious climbers. Musculoskeletal disorders such as muscle shortenings of the forearm flexors or a climber's back (humpback) must also be looked for. Highly intensive finger board and campus board training should not be performed

in these age groups. While some youth climbing teams may have medical supervision, very few adult climbers do [2].

The international standards for these team examinations are recommended by the International Sports Climbing Federation (IFSC) Medical Commission in 1999 [3] and include:

- Sports-medical examination (morphology climbing including weight, height, BMI, plicometry (body fat), flexibility, lung and heart auscultation, skin examination)
- Standard Laboratory test (blood cell count, as indicated by the clinical examination)
- ECG (12 channels) standard and stress test (either step test or bicycle)
- Spiroergometry (bicycle or step test)
- Echocardiography (first visit of a new team member, then at least every two years)
- Orthopedic examination (joints, posture, muscular dysbalance)
- Further examinations (X-ray, MRI, Ultrasound) as medically necessary (by IFSC MedCom, Paris 2009 [3])

Even though the IFSC MedCom recommends yearly examinations, they are not mandatory for World Cup athletes in this form. The representing federations only need to certify that a medical examination has been performed. In Germany, we have been examining our national team athletes according to the standards listed above, every year for the last 20 years. Additionally, all athletes' fingers received an ultrasound examination to check for injuries (e.g., tenosynovitis, pulley injury, ganglion cysts, etc.) in adult athletes and warning signs of growth plate fractures in adolescent athletes [4, 5]. In local, regional, or state teams, a stepwise reduction of these examinations is performed. The extent of the examinations also depends on the financial means of the respective team. Nevertheless, a physical examination, patient history, finger ultrasound, rest ECG, a laboratory test (blood count, urea, creatinine, electrolytes, cholesterol, triglycerides, iron, ferritin, magnesium, and blood clotting parameters, in accordance to the recommendations of the German Olympic Board), and plicometry (using the Jackson-Pollock skin measurements [6]) for body fat detection are always performed. All athletes answer a questionnaire with regard to injuries and training pat-

terns. In addition to the medical and family history, the questionnaire contains questions about sleep disorders, eating behaviors, vaccination status, and general risk factors, e.g., alcohol or nicotine consumption, allergies, medications, and supplements. We always inform the athletes about anti-doping and prohibited substances and how to proceed if medication is necessary [7].

19.2.1 Orthopedic Aspects

We focus specifically on the general and athletic history, physical examination, and finger ultrasound [4, 8]. In prior analyses, we found a history of swollen finger joints in up to 42% and a minor decreased range of motion (extension deficit $>5^\circ$ in the proximal interphalangeal finger joint) in 32% of Youth National Team Members [9]. Fortunately, these problems are not as prevalent anymore; recently, we had only 7.5% [3] climbers (40 climbers of the German national team over 2 years) with mild finger contractures [1].

The ultrasound examination is used to see if pathologies are already detectable and to determine if the growth plates have started to close [1, 4]. Special attention is given to the time frame during which the growth plates start closing, visible through a palmar-sided closure of the plate while still open on the dorsal side. As the growth plate is more prone to injury in this time frame [4], special information is given to the young climbers and their parents to look out for pain at the dorsal aspect of the finger under and after stress, if they are in the growth phase of finger middle phalanx growth plate closure [1].

In our recent analysis of the German national team examinations of 2016/2017, the results of 17 girls and 23 boys were presented [1]. The mean age was 17.5 years (18.3 years for the boys and 17.4 years for the girls), and they had been climbing for 10.9 years (the boys for 11.3 and the girls for 10.3 years). Typical findings were proneness to infections, musculoskeletal injuries, and, in one case, severe headaches. Thirty-six percent of our athletes had to take a break from climbing due to injury or illness. Most of these breaks were due to infections (7 in total), mainly minor respi-

ratory or gastrointestinal, with one severe case of pneumonia during a World Cup in Japan. The rest had to take a break due to finger injuries [2], foot injuries [2], and injuries to the shoulder and biceps [1, 2]. There was also one case of biceps tendonitis and one case of tibialis anterior syndrome. One climber needed arthroscopic surgery for a CAM impingement of the hip.

19.2.2 Pediatric and Medical Aspects

It is also important to consider the fact that in children and adolescents, the team doctor may be the first doctor these young athletes have seen since their early childhood. Therefore, it is important to check their vaccination schedule, as overseas travel may require further immunizations, or their immunizations could be incomplete (Schöffl et al., 2019), which raises the debate as to whether vaccinations should be mandatory for high-level athletes competing in Olympic sports [10]. Certain vaccinations such as Hepatitis B are recommended internationally in all athletes [10] and should be present in all athletes. For a thorough history, the HEADSS (home, education, activities, depression, drugs, suicide, and sex) mnemonic can be used in order to assess the patient's psychological status. In the past, inquiring into these fundamental aspects has allowed us to detect several pathologies. Cases of burnout, anorexia athletica, and bulimia, or, in other words, chronic energy deficiency in sport (RED-S), were especially suspected. In these cases, a psychological or psychiatric evaluation has to be initiated. If anorexia athletica is suspected, an MRI of the head is performed to rule out intracranial tumors, and, if indicated, an extensive pediatric examination with additional hormonal workup is carried out. In addition to the physical examination, a detailed analysis by a child psychiatrist is essential for evaluating the mental health of the athlete. In a study investigating the annual German team examination in the years 2017 and 2018, there were no athletes with a suspected anorexia athletica (Schöffl et al., 2019).

There was one case of a suspected immunodeficiency, which was later discarded, but a high

susceptibility to infections during the competition period in all athletes (36% of all athletes had to take a break from training or competitions during the season) (Schöffl et al., 2019). One athlete even contracted severe pneumonia while in Japan for a World Cup and had to be hospitalized on site (Schöffl et al., 2019). Two girls received further workup due to secondary amenorrhea and therapy-resistant headaches, respectively. One athlete presented with an iron deficiency anemia, and one athlete presented with hyperkalemia but showed no symptoms (Schöffl et al., 2019).

Another important function of the yearly team examination is the education of the athletes with regard to doping and permissible and forbidden use of medications. It needs to be stressed that there are “white lists” provided by national anti-doping agencies with detailed explanations of supplements and medications. The athletes need to understand that the use of supplements and medications is their responsibility and that every substance must be checked before being applied internally or externally. Most positive doping results in sports are a consequence of dietary mistakes and are based in the unknowing intake of forbidden substances.

19.2.3 Cardiac Aspects

Even though there is little doubt that physical activity has positive effects on reducing the risk of cardiovascular disease [10], elite athletes train between 10 and 20 h per week at high intensity. These high training loads could potentially lead to adverse adaptations resulting in sudden cardiac death (SCD), or an underlying cardiovascular abnormality could lead to SCD as a consequence of an exercise-induced catecholamine surge acting on an arrhythmogenic substrate [11].

The resting and stress ECG and echocardiography are standard sports-medical examinations required once a year [12]. While these are established standards for Olympic sports, they are also recommended by the IFSC Medical Commission for climbing athletes [3, 13].

Echocardiography is vital for the detection of dangerous conditions such as long QT syndrome, myocardial hypertrophy, serious arrhythmias, myo- or pericarditis, and Brugada syndrome [14]. Even though these conditions are rare, it is necessary to exclude them as possible diagnoses. Adolescent athletes present a greater prevalence of training-related and unrelated ECG changes than non-athletes such as a longer PR interval, greater frequency of sinus bradycardia, first-degree AV block, incomplete right bundle branch block (IRBBB) and voltage criteria for left ventricular hypertrophy, and early repolarization without representing a pathology [14]. Even though its use for detecting athletes at risk of SCD is controversial, the ECG can be used for identifying adaptations such as left ventricular hypertrophy (LVH), the detection of dangerous arrhythmias (WPW and long-QT syndrome) and Brugada syndrome [14].

However, most cardiac structural changes are missed using ECG, and this is where echocardiography becomes essential. Hypertrophic cardiomyopathy is the leading cause of SCD in young athletes, followed by coronary anomalies [15], both of which can be diagnosed using echocardiography. However, HCM can be especially difficult to differentiate from left ventricular hypertrophy (LVH) often seen in athletes, as measurements can overlap. Key parameters for investigating the adolescent heart using echocardiography and ruling out cardiomyopathy are left ventricular end-diastolic diameter (LVEDD), maximal septal thickness (IVSD), left ventricular wall thickness (LVPW), the dimension of the left atrium, and the transmitral E/A ratio [16, 17]. The adaptations of the heart to athletic activity vary according to the type of sport. Dynamic (endurance) sports are characterized by a high amount of movement and only a minimal amount of force, which leads to increases in heart rate and cardiac output. This in turn means that dynamic sports cause a volume load on the heart, which leads to an enlargement of the left ventricular internal diameter and a proportional increase of wall thickness, termed eccentric LVH [18]. Static, or so-called resistance, sports (weightlifting, martial arts, field throwing) involve minimal

movement but a high amount of force [19], which leads to an elevation of cardiac output as a consequence of an increase in heart rate and blood pressure. This pressure load on the heart leads to a thickening of the ventricular wall but unchanged internal dimension, termed concentric left ventricular hypertrophy [20]. A study investigating the effects of elite climbing on the heart of athletes from the German Junior National Team showed adaptations more in line with concentric left ventricular hypertrophy, so that a classification as a static sport could be suggested [14]. Interestingly, the dimensions of the left ventricular wall were comparable to those of an age-matched control of athletes from the German Junior National Nordic Skiing Team, suggesting that the adaptations of elite climbing on the athlete's heart are noticeable [14].

These observations were strengthened by the fact that the cardiopulmonary exercise results of these athletes were also comparable to athletes in team and combat sports and inferior to endurance athletes. Cardiopulmonary exercise testing (CPET) is usually implemented to determine endurance performance and cardiopulmonary fitness in elite athletes. The most important parameter studied using CPET is the VO_2 peak. Power athletes have lower VO_2 peak than team and endurance athletes [20, 21]. Thus, evaluating VO_2 peak during a standard cardiopulmonary exercise test could allow for a classification of climbing. In a study which compared elite climbers with elite Nordic skiers using a cycle exercise test, the climbers performed much worse than the Nordic skiers but were comparable to team and combat athletes [22]. Furthermore, CPET is an important instrument for assessing the risk of sudden cardiac death in athletes [23, 24] especially when employed in combination with an ECG. It allows for differentiation between physiologic left ventricular hypertrophy and hypertrophic cardiomyopathy [25, 26], one of the main reasons for SCD in athletes and difficult to diagnose using only echocardiography and ECG. Finally, CPET can be used to individualize training programs for athletes. Aerobic capacity has been shown to be important for climbing-specific recovery [24, 27] (Fig. 19.1).



Fig. 19.1 Spiroergometry during the yearly team exam

19.3 Climbing Competition Care

With the growth in the climbing gym business, sport climbing competitions have seen a dramatic increase in popularity. In addition to the large events such as World Cups, National Championships, and Masters (e.g., Adidas Rockstars, Arco Rockmasters), smaller competitions are held on state and regional levels, as well as city championships and fun competitions in almost every commercial climbing gym.

For the larger events, a medical doctor should be serving on site as a competition medical officer; in the smaller events, paramedics normally cover the competitions. The demands and regulations vary between various countries, based on the respective legal situation. For all IFSC events, a medical doctor must be on site from the opening of the isolation until the end of the competition. The following sections we will present some aspects and considerations for the medical officer of a climbing event.

Most injuries which require treatment during a climbing event are minor. In lead climbing, rope burns, skin bruising, and the occasional “wall rash” from rubbing against the rough texture are the most common ailments of competitors [28]. More severely graded trauma, e.g., dislocated ankle fracture based on a ground fall or polytraumatic injuries, are rare [1]; nevertheless, if they occur, they demand goal-oriented action, professional skills, and the necessary equipment (Fig. 19.2). Higher graded injuries are



Fig. 19.2 On-site (Boulder World Cup) treatment of a knee injury



Fig. 19.3 Taking care of athletes during the World Championships

more likely in bouldering events than in lead or speed climbing [2, 28]. As in other sport events, the medical officer is faced with a different setting than the normal working environment. In a hospital ER setting and as an emergency physician in pre-hospital care, we always work in teams with nurses, paramedics, and others. If a major incident at a climbing competition happens, the medical officer is, at least initially, alone. This is complicated by the environment, in which speakers are blaring music, access to and from the base of the wall is often blocked by competitors, and the officer is often literally in the spotlight as a crowd of people watch to see what happens. To alleviate some of the stress, there are a few questions medical personnel should have answered before the competition begins [29].

“Questions to be clarified before the competition begins:

What are the medical facilities and potential circumstances at the competition site (e.g., medical room, EMT equipment, another emergency physician at site, etc.)?

How and who do I call to get an ambulance to the site (phone numbers, EMT stations)?

Can an ambulance or gurney get to the base of the wall?

What are my communication systems? (radio, mobile phone numbers, speakers, microphone)

What other people working on site have medical training and can be of help?” [30–32]

If there is another medical service on site (e.g., mountain rescue, paramedics), contact should be established before the beginning of the event.

This helps to reduce “friction” in the event of a real emergency. Radio or cell phone contact with these other services should be established, as well as with the head of the jury and the competition’s delegate [2]. The medical area at the competition must be defined and marked. Access to the wall and road access must be clarified, as well as knowledge about the hospital environment around the competition site. The role of the medical officer must be clearly defined: is he or she “just” in charge of the competitors or does he or she serve as a doctor for the whole event, including all spectators? This can hardly be guaranteed by one person who, as the competition officer, needs to be always close to the wall, because if things happen, they happen fast. Being in charge of the whole event in the Munich World Cup with three doctors, we have already treated non-climbing conditions such as sunburns, heatstroke, heart attack, and a miscarriage. Nevertheless, the athletes themselves often complain about non-climbing-related conditions, e.g., headaches, a cold, a runny nose, etc. (Figs. 19.3 and 19.4.). With respect to all of these aspects, attention must be paid to the WADA positive and prohibited lists. In general, we only have medication in our kits which is WADA approved. This excludes our emergency kit, which is only used in a real medical emergency. Make sure to note the yearly changes of the WADA lists. In Germany, an annual anti-doping course is mandatory for all national team physicians. We strictly confirm the “no needle policy” of the IOC. It is not the job of a competition’s medical officer to inject athletes



Fig. 19.4 Acute treatment of a finger injury

with substances so that they are pain-free and can compete! The doping control system per se has nothing to do with the medical officer and is purely handled by an independent third party. At World Cup events, the IFSC Medical Commission has an injury surveillance system that requires any and all injuries to be reported. An IFSC delegate is supposed to collect these reports from the medical officer and submit them to the IFSC for further analysis.

The IFSC has serious regulations about participating in an event with a bleeding wound. This includes “flappers” on the fingers, for which an athlete can be disqualified if the bleeding is not brought under control (Fig. 19.5). Bouldering events, during which knee bars and heel hooks are common, have a lot of serious scratches that are technically “bleeding events” and thus must be attended by a physician. In the last few competitions at which we have worked, our doctors have been quite busy controlling such bleedings. The same move attempted by numerous boulderers can often create the same injury, competitor after competitor, and make for a hectic afternoon for the medical personnel. One method for dealing with bleeding scrapes or cuts is to clean the wound and then swab it with vasoconstrictive nose drops (e.g., oxymetazoline or epinephrine) to stop the bleeding before applying a bandage. Epinephrine seems to work the fastest and is the widely accepted bleeding medication in boxing events. Remember that epinephrine is on the WADA list and is only allowed as a topical medi-



Fig. 19.5 Skin bleeds on the fingers need to be stopped and covered before a climber can continue to compete

cal drug (dental, nasal, or ophthalmological) during competitions and not intramuscular. Therefore, we generally try not to use it at all [2]. Blood stains on holds need to be removed before the next climber can compete to avoid blood-to-blood transmission of HIV or hepatitis [33].

The organizers and jury committee are responsible for the safety of the athletes during an event. However, if the medical officer on site sees a problem (a split between bouldering mats, large runout between bolts, etc.), he or she can and should protest the problem. If this is done with some tact and manners, it is generally accepted and even appreciated [2].

19.4 Conclusion

The annual team examination is an important tool for prevention and early detection of orthopedic and/or medical conditions. It also helps to establish an element of trust between the athlete and the team doctor. Early signs of overstrain injuries, e.g., epiphyseal growth plate injuries, can be detected, and further harm can be prevented. The internal medical and pediatric components should not be underestimated, as recurrent infections or an incomplete state of vaccination is frequent. A cardiologic evaluation is also important to minimize the risk of sudden cardiac death in athletes. While the work on competition site is mainly treating acute injuries and disorders, the yearly exams are important for prevention and securing the athlete's

future, both as an athlete and as a healthy person. Our main goal in working with athletes is not the success of the athlete but the prevention of sport-related harm and the preservation of health. We are happy to see our athletes become world champions, but our goal as team doctors is to give them the care that they will continue to receive after completion of their athletic career: a healthy future without long-term medical consequences. Our patients should be able to look back at their time as a professional athlete with delight and look forward to a prosperous future.

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